



Advances in Automotive Electronics

Monitoring Teen Driving

Handing over the keys to the family car is a leap of faith that many parents make with their newly licensed teen drivers. In winter 2015, the 2016 Chevrolet Malibu went on sale with General Motors' (GM's) first application of its new Teen Driver feature that should give parents a bit more peace of mind. After all, it is a factory-installed hard-wired feature that cannot be easily disabled (unlike aftermarket devices that plug into onboard diagnostic ports), and it is not an application that teens can turn on and off almost at will [1], [2].

Teen Driver is an optional feature on the 2016 Chevrolet Malibu mid-level LT model, and it will be a standard feature on the Malibu Premier model. The system needs to be activated after purchase, but once a parent turns it on, it stays on and requires a four-digit personal identification number code to disable it, therein erasing all history and clearing the screen [2].

The Teen Driver feature allows parents to set speed alerts, limit audio volume, and receive vehicle operating reports. A spokesman for

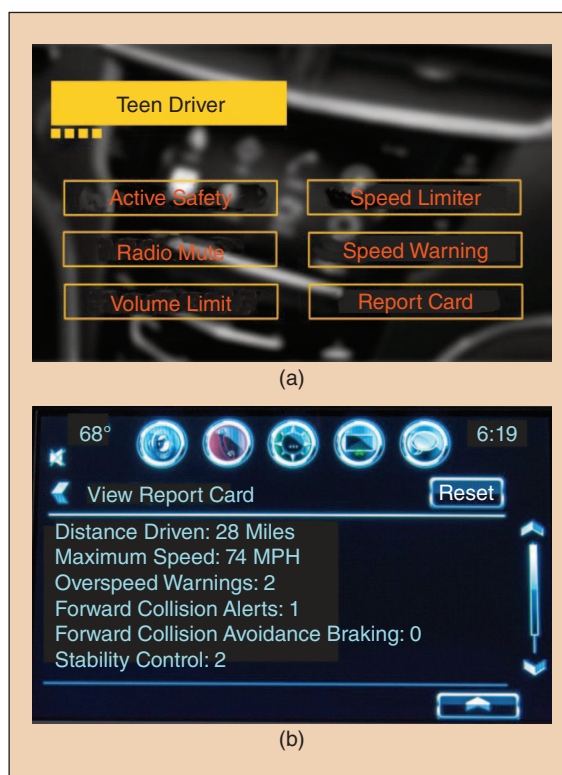


FIGURE 1 (a) Some of the limits on the vehicle operation that parents can set using the Teen Driver option. (b) The display of a customized report of events that occurred after a teen had been driving [1]. (Photos courtesy of Chevrolet.)

Chevrolet said, "Parents could use it as a teaching tool with their kids, and they can discuss and reinforce safe driving habits" [1].

Teen Driver will not prevent your child from driving above the speed limit, but it will record the distance traveled, the highest speed reached, and the number of exceptions to speed parameters. It also monitors

the triggering of safety features, such as [2]

- antilock brakes
- electronic stability control
- forward-collision alert
- lane-departure warnings (which often are a telltale sign of distracted driving).

Teen Driver also prevents parent-selected parameters from being disabled. Its speed warnings flash when the driver exceeds a preset limit, adjustable from 40 to 75 mi/h (64–121 km/h). Parents can also set sound-system volume limits [1].

Using inputs like those shown in Figure 1(a), parents can create customizable reports that include such things as distance driven, top speed achieved, preset speed warnings exceeded, stability-control events, antilock brake events, forward-collision alerts, and autobraking events. A Teen Driver customized report display, showing events that occurred when a teen was driving, is shown in Figure 1(b).

In addition, parents can dictate what features can or cannot be deactivated. Then, control over the activation status of stability control, parking sensors, blind-spot monitoring, rear cross-traffic alert, forward-collision warning, automatic braking, and traction control can all be set so they cannot be altered by the teen driver [1].

The Teen Driver option does not directly address one of the most alarming trends among younger drivers these days: distraction via mobile devices, especially smartphones. Chevrolet noted that a teen driver in-car device-muting technology—which disables phone use when the car is moving—is being worked on in the industry as a whole, but it still is not ready for production [1].

Nevertheless, there are ways teens might get around Chevrolet's best intentions [2]:

- The Teen Report Card tracks only driving associated with a specified key fob. When your teen drives the car with his or her key, it records data. When you drive the car with your set of keys, the car will not record anything. Parents might need to lock up their spare key and maybe also paint the plastic fob part of the teen's key a florescent yellow to better see that the teen uses his or her correct key.
- If parents want to know how far or where their teen is driving, they have to clear the Teen Driver Report Card on a daily or weekly basis, monitor the distance driven, and try to calculate where the car has been. If teen drivers learn the pass code, they could just clear the data ahead of time and pretend they never left the house.
- By design, Teen Driver does not time-stamp data (to prevent the data from being used by insurance companies to review claims). As an example, a lane-departure warning alert seen on the screen could have happened yesterday, or it could have happened last week or last month.
- Teen Driver mutes the vehicle's (or any paired/connected device's) audio if it detects that the driver's seat belt is not buckled, which is great since teens sometimes resist buckling up. On the other hand, Teen Driver does not

track whether or not other passengers are in the car. Parents will not know if teens are abiding by new laws that limit the number of passengers in the car (which research says can be a major source of distraction).

Chevrolet realizes that Teen Driver is no panacea. "It's meant to be used as a teaching tool. It's why Chevrolet calls it a report card. So it's best to look at this feature as an educational tool rather than a solution," said Chevrolet spokesperson Chad Lyons [2].

Variable Cluster Display

Audi AG (Ingolstadt, Germany) is featuring an innovative instrument cluster in front of the driver in its new 2016 TT sports coupe (shown in Figure 2). In the instrument-cluster space, an all-digital video display is located so that it can project multiple different screen displays of selected information [3].

An automotive journalist commented, "It was a far cleaner design than other display screens automakers typically mount in the center stack of the dashboard," adding, "I did not find it distracting" [3].

Audi calls its video instrument cluster a "virtual cockpit." It is all digital, and it is powered by an embedded visual computing module (VCM). The VCM includes an Nvidia quad-core Tegra 3 processor. Tegra 3 is a system-on-chip processor developed for mobile devices. It provides fast, 60-frame/s response video display for the Audi 12.3-in (31.2-cm) display screen. The screen provides



FIGURE 2 Audi's 2016TT sport coupe features an all-digital video instrument cluster [4]. (Photo courtesy of Audi.)

1,440-pixel $\times$ 540-pixel high-resolution images. The fast response time helps the digital display readouts such as the speedometer and tachometer to react in real time as the car is driving [4].

Audi's "virtual cockpit" man-machine interface (MMI) display may sound gimmicky to some driving purists, but everything from the classic tachometer and speedometer to Google Earth three-dimensional (3-D) maps were reported as "looking great on this display."

A driver can select various MMI display screens using a steering wheel control button. So, to operate the display, you do not have to glance down to the usual rotary and touch-pad controls on the center console, but these controls are still there if you (or your passenger) need them.

Various screen displays, selectable in the virtual cockpit, include the following [4]:

- The classic view, shown in Figure 3(a), with the tachometer and speedometer in the forefront: the smaller images for fuel, engine temperature, and other basic information run along the top and bottom.
- What drivers often want in front of them is a map, as shown in Figure 3(b). The data are provided by a digital mapping service (recently purchased by a consortium of automakers, including Audi). There is also a Google Earth 3-D visual overlay to make the map more visually accurate and lifelike.
- An infotainment view (not shown), where the tachometer and speedometer are again shrunk to allow more space for infotainment information, is also available.
- A second infotainment view (not shown) displays how weather and other important data can come forward when you want it. Remember, this is a digital display, so you can change the view whenever you wish.
- Another option is a Google Earth view (not shown) where the open road in front is displayed almost

like looking out the windshield twice. It shows off the quality of the image data available [4].

The internal components of the virtual cockpit are shown in Figure 3(c). This sectional view of the virtual cockpit module provides a look at the computer inside it along with the special glare-reducing, thin-film transistor-driven display. One of Nvidia's VCM circuit boards is shown in the lower portion of the shown photo. There is a second VCM (not seen here) that controls just the navigation screen view [4].

Advanced Driver-Assistance Systems

Advanced driver-assistance systems (ADASs) can be defined as follows [5]:

- ADASs enhance vehicle safety by helping a driver to avoid collisions through new technology implementations, in some cases taking over control of the vehicle.
- ADASs can help to automate lighting and provide adaptive cruise control. They also automate braking, incorporate global positioning system/traffic warnings, connect to smartphones, alert the driver to other cars or dangers, keep the driver in the correct lane, and show what is in blind spots.
- ADAS technology can be based on vision/camera systems, sensor technologies, car data networks, vehicle-to-vehicle (V2V) systems, and vehicle-to-infrastructure (V2I) systems.
- Next-generation ADASs will increasingly leverage wireless network connectivity to offer improved value using V2V and V2I data, which lead to partial or full automation of the driving experience [5].

Making Automatic Braking Standard

Automatic emergency braking (AEB) is an example of an ADAS. Ten major automakers have reached an agreement to add AEB as a standard feature on all their new cars. It is

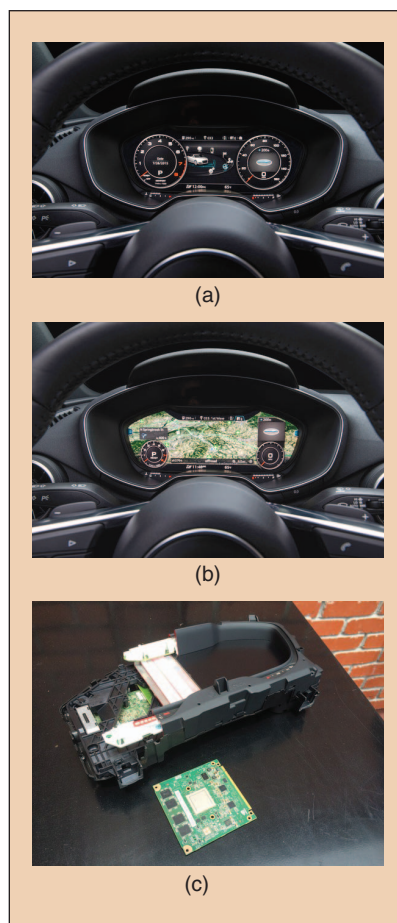


FIGURE 3 (a) The virtual cockpit's classic display view. (b) The digital road map view with the tachometer and speedometer shrunk to provide more space for the road map. (c) The cut-away section of the module. The Nvidia VCM circuit board is shown in the lower portion of this photo [4]. (Photos courtesy of Audi.)

predicated that AEB systems could reduce auto insurance claims by as much as 35%. Additionally, the U.S. National Transportation Safety Board issued a 60-page report in June 2015 asking that automatic braking be required on new cars, saying it would save 1,700 lives a year and eliminate 500,000 injuries [6].

An AEB agreement was announced on 11 September 2015 by the U.S. Department of Transportation (DOT), its National Highway Traffic Safety Administration (NHTSA), and the autosafety arm of the U.S. insurance industry, the Insurance Institute for Highway Safety (IIHS).

This announcement represents a major step toward making crash-

prevention technologies more widely available to consumers. The ten companies—Audi, BMW, Ford, GM, Mazda, Mercedes-Benz, Tesla, Toyota, Volkswagen, and Volvo—will work with the IIHS and NHTSA in the coming months on the details of implementing their historic commitment, including a time line for making AEB a standard feature. This agreement also shows that the U.S. government is willing to work to forge agreements with automakers rather than simply mandating safety improvements [6], [7].

“We are entering a new era of vehicle safety, focused on preventing crashes from ever occurring, rather than just protecting occupants when crashes happen,” said DOT Secretary Anthony Foxx. “But if technologies such as automatic emergency braking are only available as options or on the most expensive models, too few Americans will see the benefits of this new era” [7].

The ten manufacturers committing to across-the-board AEB systems represented 57% of U.S. light-duty vehicle sales in 2014. The DOT and IIHS encourage all other light-vehicle and trucking manufacturers to bring automated vehicle technology to all vehicles on U.S. roadways as soon as possible.

AEB is already in some mid-market vehicles, usually as a US\$250–400 option, but this new agreement would ultimately bring it to every vehicle. It could represent the next big safety advancement that could have the impact of seat belts, antilock brakes, and stability control [6].

AEB includes a range of systems designed to address the large number of crashes, especially rear-end crashes, in which drivers do not apply the brakes or fail to apply sufficient braking power to avoid or mitigate a crash. As shown in Figure 4, AEB systems—using vehicle sensors such as radar, lidar, and/or cameras to detect an imminent crash—warn the driver, and if the driver does not take sufficient action, they automatically apply the brakes [7].

“The evidence is mounting that AEB is making a difference,” said IIHS President Adrian Lund. “Most crashes involve driver error. This technology can compensate for the mistakes drivers make because these systems are always on alert, monitoring the road ahead and never getting tired or distracted” [7].

“Forward-collision warning with automatic emergency braking is the biggest safety advancement since the introduction of stability control over two decades ago,” says Jake Fisher, director of auto testing for Consumer Reports, which issued a statement supporting the move. “This is such an important safety feature that all manufacturers should bring it to their vehicles as soon as possible” [6].

Moving forward, IIHS and NHTSA will set specific performance criteria for manufacturers to meet their commitment and will determine how soon consumers can expect to see AEB technology as standard equipment. For a vehicle to earn IIHS’s highest safety award, Top Safety Pick, IIHS will require that a vehicle has an automatic braking system [7]. IIHS introduced its AEB crash-avoidance/mitigation test in 2013. The AEB test procedure is described in [9].

Low-Cost Lidar

AEB systems utilize on-vehicle sensors, generally including lidar (light detection and ranging), to help detect and identify an imminent crash with a vehicle up ahead.



FIGURE 4 An AEB system detects an imminent crash and warns the driver, and if the driver does not take sufficient action, it applies the brakes [8]. (Photo courtesy of Volvo Car Corporation, Gothenburg, Sweden.)

OSRAM MAKES THE LIDAR LASER AND PHOTODIODE COMPONENTS COMPOSED OF ULTRAFAST-RESPONSE INDIUM–GALLIUM–ARSENIDE SEMICONDUCTORS.

The cost of a lidar sensor is expected to eventually come down, well under US\$100. “One low-cost lidar under development utilizes a pulsed laser along with a two-by-eight array of 16 photodiodes, and it has a detection range of 30 m,” says Rajeev Thakur, a marketing manager for Osram Opto Semiconductors, Munich, Germany. The photodiode array is wider than it is high because horizontal resolution is what matters most for the intended application, AEB systems [10].

Osram makes the lidar laser and photodiode components composed of ultrafast-response indium–gallium–arsenide semiconductors. Its partner, the start-up company Phantom Intelligence, Quebec, Canada, integrates Osram’s components into the lidar’s electronics module, which has a footprint about the size of a business card. The full sensor package (shown in Figure 5) is larger, but it is claimed that it can fit on the inside of the windshield, just ahead of the rear-view mirror.

The lidar sensor neither scans mechanically, like a roof-mounted one, nor electronically. Instead, it emits near-infrared (IR) light from a dedicated IR-pulsed laser diode. The 16 photodiodes detect pulses reflected in two vertical planes, with each plane segmented into eight horizontally spaced IR-photodiode detection channels. The received IR pulses are digitally processed to compute the distance and the rate of change in distance as detected in each of the 16 channels. The photodiodes are especially suitable for automotive lidar applications for their given fast switching time [11].

The lidar’s electronics module is being developed by Phantom Intelligence. Osram provides the IR-pulsed laser diode and the photodiodes.

The laser delivers a high efficiency of up to 40% at 70-W peak output power. The laser, however, emits very short 30-ns pulses that together are only activated 0.5% of the time, making the laser beam safe for the eyes of passersby’s, while also providing a long operating lifetime [10], [11].

Because the lidar sensor uses a single IR laser and off-the-shelf photodiodes, it is significantly lower cost than, for example, an array of microwave radar antennas. The sensor has been proven reliable for low-speed obstacle detection in city driving [10], [11].

Thakur noted, “Because new regulatory policies in Europe will grade new cars on the AEB feature by 2018, the low-cost lidar sensor could in the near future find a large market there. In 2018, European safety regulators will start grading new cars on how well they can avoid colliding with a car ahead of them in slow-moving traffic—precisely the kind of thing that most tends to lull drivers into inattentiveness. Although no minimum Euro test-performance grade will be mandatory, a good grade will constitute an important marketing incentive. Europe has even tougher targets on its industry road map for the year 2020, like recognizing

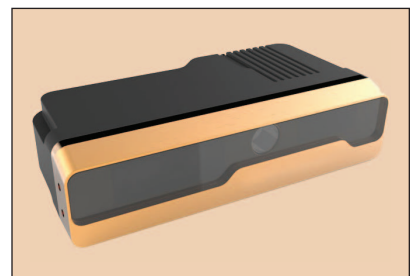


FIGURE 5 A low-cost lidar sensor for use in AEB systems. The sensor will fit on the inside of the windshield [10]. (Photo courtesy of Osram.)

GM MAKES SURE THAT CARPLAY MEETS BOTH ITS OWN SAFETY EXPECTATIONS AND THOSE OF NHTSA.

children on the sidewalk as well as pedestrians on the roadway” [10].

“Additionally, the U.S. regulator, NHTSA, is working on a slightly more ambitious grading system for 2019. For example, it is also looking at things like cross-traffic alerts. In this case, you might need to combine two sensors, like lidar and radar,” said Thakur.

Osram and Phantom Intelligence are also talking with potential customers about integrating the lidar sensor in headlamps, which are designed to let light through, while windshields often have special coatings to block out certain optical frequencies—similar to those in the Osram lidar IR spectrum. “True, dirt would tend to accumulate on the headlamp, but that problem is solvable,” Thakur says [10].

Smartphone Connectivity

GM is the first automaker to roll out both Apple CarPlay and Google Android Auto across nearly its entire Chevrolet product line. Apple CarPlay connects Apple’s iPhone to the vehicle, and Google Android Auto connects its Android mobile operating system to a vehicle [12].

Neither Android Auto nor Apple CarPlay systems can be retrofitted into pre-2016 GM model cars. The Google and Apple systems also will

not replace either Chevrolet’s existing MyLink infotainment system, which drivers will still use for vehicle diagnostics, or OnStar (GM’s subscription-based vehicle communications system), which offers services such as roadside assistance [13].

You might think GM would not want to partner with monolithic companies like Apple and Google who seemingly are intent on world domination. But customers want an in-car system that operates as seamlessly and intuitively as the smartphones do themselves [12].

Apple and Google Partner with GM

For 20 years, GM has been at the vanguard of telematics (the long-distance transmission of computerized information), infotainment, and in-car connectivity. GM launched OnStar at the 1995 Chicago, Illinois, Auto Show. GM’s OnStar has had over 1 billion interactions. It originated hands-free calling, navigation, roadside assistance, and other diagnostics.

Chevrolet’s MyLink voice recognition system and the similar Intellilink system (offered on other GM cars) represent an evolution of OnStar. Currently, OnStar connects voice commands, navigation, radio, climate controls, and other vehicle diagnostics to car occupants. In short, you can control your creature comforts from a screen. “Apple CarPlay and Android Auto provide enhancements to MyLink, not a replacement,” explained Terry Inch, chief operating officer for GM’s OnStar operation [12].

How It Works

Plug in your Apple iPhone 5 or later smartphone to the car’s universal serial bus (USB) port, and an Apple CarPlay icon appears on the phone’s screen. The GM vehicle must be

equipped with a MyLink 7- or 9-in (18- or 23-cm) center-console screen. Certain base-level car models will not support smartphone connectivity. Click the large icon that first appears on the dashboard screen, and, as shown in Figure 6, you will get a condensed version of what was on the iPhone screen, with two rows of four large, easy-to-access touch-input icons. Nothing is jammed in the corners. It is not projection, but integration. Next, stow the phone because you are locked out of it as long as it is plugged into CarPlay, and, besides, the phone’s screen also goes blank [12], [13].

Not every application on a phone will transfer to the screen. For example, any application such as YouTube that opens up a streaming video service will be blocked for safety purposes. GM, working with agreements of both Apple and Android, makes sure that CarPlay meets both its own safety expectations and those of the NHTSA. For example, you cannot play the video game *Angry Birds*. You also cannot customize and add your own applications. You get a select number of applications you can choose from based on what Apple and GM will offer [12], [13].

As shown in Figure 7, when Apple’s CarPlay is connected, everything on the dashboard display screen can be controlled by any of three types of inputs [12]:

- Siri voice commands
- touch-screen buttons
- other dashboard knobs and buttons.

A 3-min video demonstrating in-car operation of CarPlay is available in [12].

Audio and Voice Commands

Audio is less an evolutionary leap than a logical extension. The benefit here is that you do not need to first click the application on the phone; you simply access it on the touch screen. Radio voice commands are similar to the fidelity in MyLink, which is pretty good by industry



FIGURE 6 When an Apple CarPlay smartphone is connected to a USB port, its phone applications are sent to the car and are integrated into the dashboard screen [12]. (Photo courtesy of Chevrolet.)

standards. However, Apple's Siri offered better voice recognition commands and was more hospitable than GM's MyLink system [12].

To avoid having the driver type in commands, GM requires that all messaging and navigation services must be accessed by voice command. Users still need to be schooled in the layered system of voice commands. Phone commands such as, "dial Chicago Tribune Autos" wouldn't work unless you were already in the phone submenu. The procedure is to hold the voice button on the steering wheel (shown in Figure 7) for about 3 s. Then Siri responds just as she would when you are pressing the call button on the iPhone. You can, for example, ask Siri to find the nearest Starbucks coffeehouse, and results will appear with a map on the in-car display [12], [13].

Text Messaging

This might be your favorite feature of CarPlay, and it is expected that e-mail read-back capabilities will soon follow. Using either voice commands or the touch screen (shown in Figure 7), you can have text messages read aloud to you and then respond via voice-recognition or simply with a phone call. If you use voice reply, Siri reads back your reply before sending, which means the grammar will likely be better than if you sent it using your thumbs on the smartphone's keyboard [12].

Cost

It does not cost anything extra to integrate the system with existing MyLink infotainment systems. You can use data from your cell phone provider or use Chevrolet's OnStar. Depending on how much data you want, prices range from US\$5 to US\$50 per month. You can pay US\$5 for one day or get 10 GB for a year for US\$150 for OnStar subscribers, or it costs US\$200 for nonsubscribers. New car buyers get the first three months free [12].

GM does not see smartphone connectivity so much as a potential

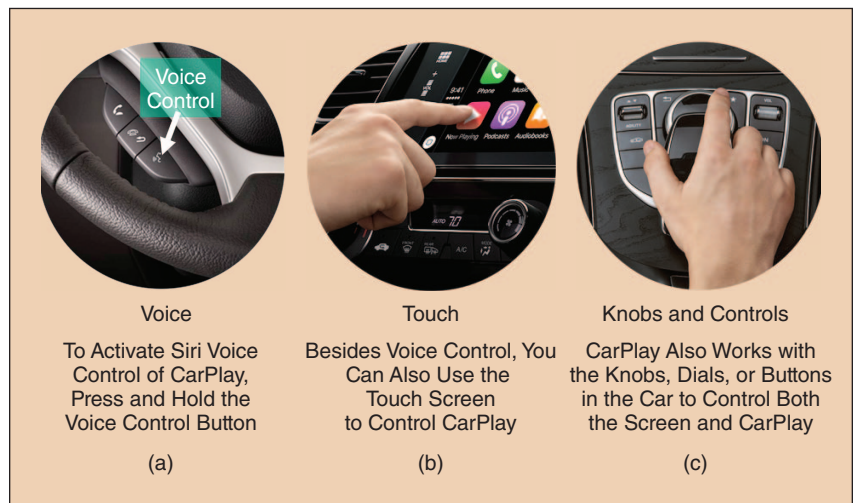


FIGURE 7 Everything on the dashboard display screen can be controlled by (a) voice, (b) touch screen, and (c) knobs and buttons [14]. (Photos courtesy of Apple.)

revenue stream but rather as a way to differentiate its vehicles from the competition [13].

V2X Communication Update

Safety proponents are looking forward to the day when cars can send each other information that can prevent accidents. V2V and also V2I communications will be used. V2I systems employ dedicated short-range communications (DSRC) and they provide information such as vehicle speeds and locations. Commonly lumped together as V2X, both technologies can play a major role in reducing accidents. Government regulators in the United States and

Europe are expected to mandate their use before long [15].

In Japan, shipments are beginning to build, giving the industry a working model. "We'll be selling V2X this year, and we've already been selling V2I (shown in Figure 8) for around four years," said Hideki Hada of Toyota Motor (Toyota City, Japan) Engineering and Manufacturing Integrated Vehicle Systems Department."

There are solid reasons for V2X interest. In the United States, the NHTSA found that V2X systems could help reduce the occurrence and severity of 22 of the 37 most-common types of crashes defined in

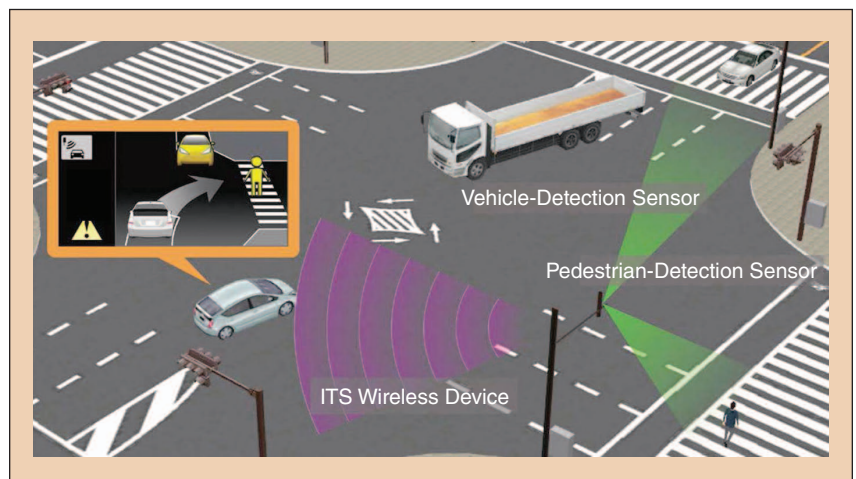


FIGURE 8 Since 2011, Toyota has continued to install V2I systems in Japan [15]. (Photo courtesy of Toyota.)

its study. That could improve safety more than many of the agency's previous efforts. "That's around 80% of all crashes," said John Capp, director, Global Vehicle Safety at GM. "If you add up all the work done on seat belts and air bags, it wouldn't match the benefits we potentially could get from V2X" [15].

However, the impact of V2X could be some years coming because, if a firm mandate comes, it will take years to build a critical mass of vehicles that can communicate with each other.

In 2015, GM highlighted its support of DSRC systems by announcing that its 2017 Cadillac CTS will have V2V capabilities. While Cadillac owners will not gain many benefits by having their cars talking to each other, GM feels that its strategy will bring a solid amount of knowledge with a small downside. If NHTSA ultimately decides to set a V2V regulation that is not fully compatible with the current GM V2V system, necessary changes are expected to be minimal [15].

V2X Cellular Communication

"If DSRC communications had come out ten years ago, it might have been the way to provide firmware updates," said Andrew Poliak, global director, business development for QNX Software Systems. (Firmware is permanent software programmed into a read-only memory.) Poliak added, "It's taken so

long to roll out DSRC that other alternatives (i.e., cellular communications) have emerged."

However, sending signals to cell towers and getting information back to vehicles introduces latency (the delay from inputting signals into a system verses desired outcome). Safety alerts cannot tolerate such delays.

There has been some talk about next-generation fifth-generation (5G) cellular techniques, which could reduce latencies to acceptable levels for safety, as a replacement for DSRC. This is because 5G systems are now being designed as a higher-performance 5G cellular technology with nearly two orders-of-magnitude faster download speeds than current fourth-generation (4G) technology [15].

Real-time communications are a prerequisite for V2V to work. In the fall of 2015, proof of V2V feasibility was established in Germany. "This is the first demonstration of V2V car-to-car communication via a high-speed cellular connection with near-5G performance," said Alexander Dobrindt, Germany's minister of transport and infrastructure [16].

The V2V experiment (Figure 9) was conducted on a recently inaugurated "digital A9 motorway test bed," a segment of an autobahn in Bavaria, Southern Germany.

The autobahn highway test bed utilized regular 4G long-term evolution (LTE) service from Deutsche Telekom (Bonn, Germany), which was upgraded locally with a Nokia Networks (Espoo, Finland) system optimized for rapidly moving vehicles. The LTE 4G service in itself offers wireless communications that provide up to ten times the speed of third-generation networks.

Nokia claims that its near-5G technology cuts total transmission lag time/latency to under 20 ms, compared with today's transmission latencies that are no faster than 100 ms. Nokia's 20-ms lag time includes the relay time from one car to another, via a central cloud computing center. It was predicted that point-to-point 5G

transmissions are ultimately expected to exhibit lag times of just a couple of milliseconds [16].

A time frame for 5G rollouts is uncertain, so coverage could be quite spotty for years. Additionally, shifting to 5G technology would require substantial testing and development work—work that has already been done for current V2X 4G systems [15].

V2X System Security

Secure communications are a key hurdle for V2X systems rollout. Manufacturers of the V2X systems will have to protect them when security information is programmed into transceiver modules. Encryption technologies in these modules will play an important role in keeping hackers from spoofing (when an intruder attempts to gain unauthorized access to a user's system) or simply trying to steal messages.

If most vehicles eventually have V2X links, the sheer size of the network will be daunting. There will not be much time for vehicles to determine whether they are communicating with a certified system or a bad actor [15].

"When a vehicle meets a vehicle it's never seen before, how do you manage trust so the vehicles both know messages are valid?" asks Mike Shulman, Ford Motor Company, Dearborn, Michigan, United States, "With over 200 million vehicles in the United States, V2X would be the largest significant public infrastructure system ever deployed."

Furthermore, security will also remain an issue over later years as vehicles need maintenance. Encryption keys and certificates must be protected even if in-car V2X modules are repaired or replaced. (A certificate authority uses a third party to identify one end or both ends of a communication to ensure secure communications between Internet servers and browsers.)

"These will be challenging security issues," Poliak said. For example, "There will be tradeoffs between security strategies versus laws which



FIGURE 9 An in-car screen displays traffic information provided by V2V communication on the "digital A9 motorway test bed" in Germany [16]. (Photo courtesy of Nokia.)

TABLE 1 Self-driving vehicles and drivers registered in California* [17].

Company	Drivers	Vehicles
Google	186	48
Volkswagen	30	3
Nissan	18	3
Tesla	16	12
Mercedes	13	5
Bosch	12	2
Delphi	9	2
Cruise Automation**	6	2

*Number of registrations through June 2015.

**Cruise Automation, San Francisco, California, is developing an aftermarket self-driving kit for Audi vehicles.

require that people are able to work on vehicles. You have to balance that right-to-work on a car versus the need to protect security systems [15].”

V2X Data Analysis

If vehicles eventually talk to each other and to infrastructure as expected, there are going to be enormous amounts of data concerning traffic and road conditions. Both information technology (IT) and automotive companies feel that some of this information has a lot of potential value, though what exactly it is good for remains uncertain.

Still, IT companies are already exploring ideas that will help them justify building large systems that focus on transportation data. There may not be one killer application for data, but many minor usage models may be developed.

“We’re making huge investments into connected cars as part of our cloud services strategy,” said Richard Nash, Solution Architect for Intelligent Transport at IBM [15].

Licensed Autonomous Vehicles

The state of California almost certainly has more autonomous (self-driving) cars and operators on public roads than the rest of the world. That California has so many of these cars and operators should not come as a surprise. For a start, it is reliably mild and dry, and the sunny climate is perfect for road-testing early generations of vehicles that would balk at snow, fog, and/or heavy rain. Moreover, virtually every technology company and most mainstream automakers have offices in California’s Bay Area [17].

But the main reason for California’s supremacy in autonomous vehicle development is the presence of a single company in Mountain View, California. While Google did not invent the self-driving car, it can lay claim to having invented the industry of self-driving cars. It purchased start-up businesses, hired experts, and developed essential mapping and navigation technologies.

From the word go, Google has usually remained one step ahead of the competition. It was the first to

- test experimental vehicles at scale (i.e., provide test results that can be scaled up to represent an entire fleet of cars)
- move from relatively safe highways to unpredictable city streets

- test a purpose-built, steering-wheel-free, and self-driving prototype.

As Table 1 shows, even now (as of June 2015) after the arrival of global carmakers and upstarts like Tesla, Google has nearly twice as many autonomous vehicles and drivers on California’s roads as all other carmakers combined.

If, as some observers suggest, Apple is also working on its own high-tech car, California will likely continue to boast the most roads with self-driving cars in the world well into the future [17].

V2V/Self-Driving Car Test Facility

Working with the state of Michigan DOT, University of Michigan (U-M) (Ann Arbor, Michigan) researchers have constructed and opened Mcity, a unique test facility for evaluating the capabilities of connected (V2V) and automated (self-driving) vehicles and systems [18]. An overview of U-M’s Mcity test facility, including photos, was published in the December 2015 issue of this magazine [19].

Located at U-M’s North Campus Research Complex, Mcity simulates the broad range of complexities that vehicles can encounter on both city streets and major highways. Mcity includes

- approximately five lane miles (8 km) of roads and a tunnel, shown in Figure 10(a)
- sidewalks and a typical city street, shown in Figure 10(b)
- fixed obstacles, such as potholes and benches.



(a)



(b)

FIGURE 10 Mcity’s V2V and self-driving-vehicle test facility. (a) It includes a tunnel. (b) It has typical city streets [18]. (Photos courtesy of the University of Michigan.)

TEGRA 3 IS A SYSTEM-ON-CHIP PROCESSOR DEVELOPED FOR MOBILE DEVICES.

New driving technologies are already being tested elsewhere. But, at the U-M test facility, a more rigorous examination can be conducted before sending cars onto public streets. This allows companies to identify how connected and automated vehicles will handle everyday road challenges, i.e., challenges that human drivers readily interpret, but are difficult for autonomous cars to assess [20].

While some companies (notably Google and Tesla) are testing self-driving cars on public roadways, mainly in California's Silicon Valley; Mcity allows companies to test and retest cars in unusual situations the cars may rarely experience on public roads. One example would be repeated tests having the same dummy pedestrian coming out from between the same two cars, both partially hidden by a bus [21].

First Mcity Test Vehicle

In November 2015, the first autonomous car to start testing at the Mcity

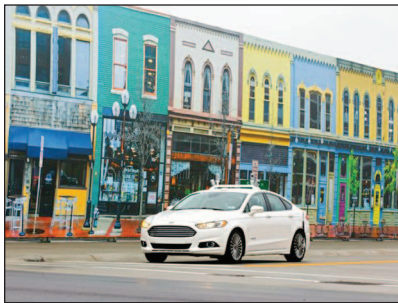


FIGURE 11 This research vehicle, a fully autonomous Ford Fusion, was the first car tested at the Mcity facility [22]. (Photo courtesy of Ford Motor.)

facility was an autonomous Ford Motor Fusion vehicle. "Testing Ford's autonomous vehicles at Mcity provides another challenging, yet safe, urban environment to repeatedly check and hone self-driving vehicle technologies," said Raj Nair, Ford Motor Group vice president, global product development [22].

As shown in Figure 11, the first Mcity test car was a Ford Fusion "hybrid autonomous research vehicle." It was equipped with cameras, radar and sensors, including lidar to generate a real-time 3-D map of the vehicle's surrounding environment. Using inputs from these sensors, the vehicle makes driving decisions as it passes through a test area complete with street lights, crosswalks, lanes, ramps, roundabouts, curbs, bike lanes, trees, hydrants, sidewalks, signs, traffic lights, and construction barriers [22].

The goal of Mcity testing is to establish vehicle scaling factors. "Every mile driven and then repeatedly redriven at Mcity can represent 10, 100, or 1,000 mi of on-road driving in terms of our ability to test the occurrences of rare, but likely, events," said Ryan Eustice, U-M associate professor and principal investigator in Ford Motor's research collaboration with the university [22].

Toyota Driverless Cars

Giant corporations in California's Silicon Valley such as Google may



FIGURE 12 Toyota's automated driving test vehicle [23]. (Photo courtesy of Toyota.)

appear to have seized the lead on self-driving car technology in recent years. But Toyota, the world's largest automaker, has recently gone public with its own plans to make some of their cars capable of autonomous highway driving by 2020. One of their

autonomous test vehicles is shown in Figure 12.

Toyota has generally taken a more cautious approach to self-driving cars by emphasizing that they will assist rather than replace drivers. The automaker named one of its prototype self-driving cars, "Highway Teammate," and, until recently, preferred to call its system "advanced driver support" rather than "automated driving."

Toyota has plans to develop a wide array of semi-autonomous driving technologies that could end up in mass-market cars within the next few years. However, "fully autonomous driving for mass-market cars may not become a reality until the costs of all the self-driving sensors, electronics and controllers go down," Toyota officials said [23].

Self-Driving Cars—Difficult Situations

Google is currently concentrating its autonomous car testing on the more



FIGURE 13 Google's car is shown here self-driving itself on a surface street in Mountain View, California [24]. (Photo courtesy of Google.)

difficult and unpredictable scenarios that crop up on surface streets more often than on freeways. Speaking at the Automated Vehicle Symposium (21–23 July 2015, Ypsilanti, Michigan), Chris Urmson, director of Google's self-driving cars, described a variety of scenarios that self-driving cars (Figure 13) have confronted on surface streets in the neighborhood of the company's Mountain View, California, headquarters [24].

Scenario 1

This case featured a duck scampering across an intersection followed by a

woman in an electric-powered wheelchair going back and forth, in apparent pursuit of the duck. “It doesn’t matter how long you gave me, I never would have come up with that scenario. There’s nothing like this in a handbook,” Urmson said.

The car must have the ability to determine what to do. “This is weird. The car was just going to wait until the situation played out.” On the other hand, human drivers might watch the path of the woman and the duck and proceed if they think they see a safe opening. However, the Google car will simply wait until its cameras and sensors do not notice any impediment [24].

Scenario 2

Urmson described another example where three lanes of traffic were about to proceed from an intersection as the light turned green. The Google car was in the far right lane. Coming from the left was a bicyclist who was determined to shoot through the intersection despite a red light in his direction.

The first two human-driven cars moved forward, forcing the cyclist to change course, turn away from the cars, and swerve left toward the middle of the intersection to avoid them. The Google car held its position because its sensors picked up the cyclist’s speed and concluded it was about to bolt through the intersection—which the cyclist did not do. But this extra degree of caution, at this stage in the technology’s development, could possibly create another level of risk, that is to say, the stationary Google car might be rear ended by a vehicle behind it [24].

Google Car Safety Experience

Google has recently begun reporting accidents involving its self-driving vehicles. It turns out that as of July 2015, its cars had been involved in 14 minor accidents over more than 1.8 million mi of testing. But Google insists that the self-driving cars did not cause one of the collisions.

“There’s been a lot of noise recently in the press about the fact that our vehicles have been in collisions,” Urmson said. He showed a video of one of the recent accidents where the Google car had to stop because of other vehicles stopped in front of it. Meanwhile, the car behind it did not notice and rammed into the rear end of the Google car.

This illustrates one of the challenges that needs to be solved before autonomous vehicles enter the mainstream of public road and highway networks. Yes, they can be programmed to be as safe as possible, given the immediate space around them. However, other drivers in non-autonomous cars are human, and it is true, “to err is human.” In other words, a self-driving car cannot, for example, prevent a human-driven car from causing a rear-end collision [24].

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